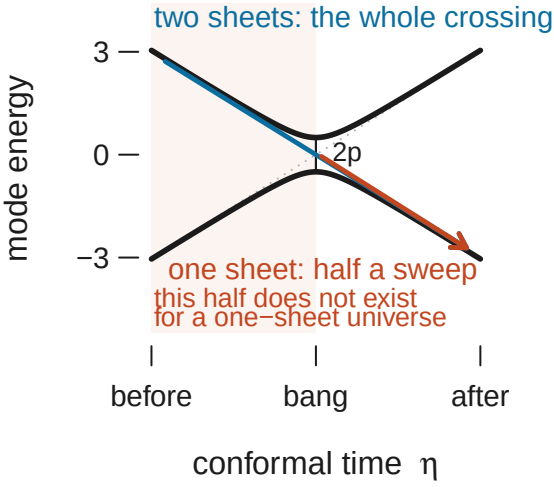
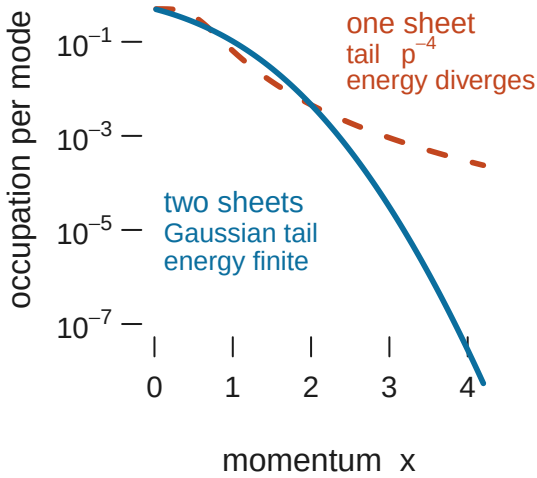
every mode crosses an avoided crossing




the two sweeps leave different states